

SNOMED CT Implementation Requirements



Health
Food Chain Safety
Environment

September 2026

NRC Belgium



AGENDA

- Goal project - Context - Purpose
- Objectives - Requirements
- Registration
- Github
- Varia

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Timeline SNOMED CT implementation roadmap

✓ **EHDS (I)**
Exchange
Patient Summary

Clinical
implementation

eHR configuration / Tooling / Education & training

3 SCT in use

Implemented in clinical
workflow

Building Content

Technical
implementation
(eHR)

SCT binding

2a SCT coded
Patient Problems

2b SCT coded
Procedures, Observ.,
Allergy, Vaccination

Foundation

1 Term server
Local TS installed
Connection NTS OK

Requirements

eHR Registration / Certification

Preparation

0 eHR requirements
published

Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4

2026

2027

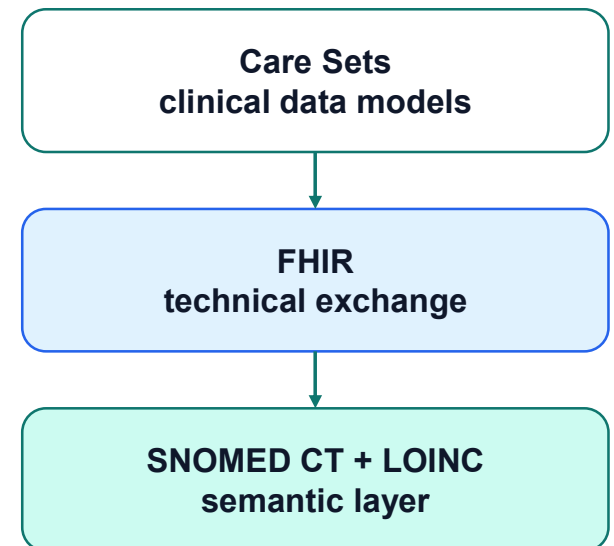
2028

2029

SNOMED CT requirements for semantic interoperability

A compact slide set based on the General section: national architecture, purpose, rationale, and governance.

Core idea: data only becomes reusable when its meaning is captured unambiguously at source.



Belgian Integrated Health Record

BIHR uses a federated model: information stays at its authentic source but is structured, coded, and semantically standardized.

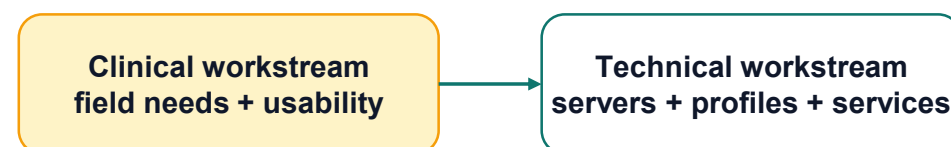
Introduction: architecture meets obligation

BIHR, EHDS, and SNOMED CT form one implementation context.

- BIHR integrates patient information without creating a new central store.
- Care Sets define what is exchanged; FHIR defines how it is exchanged.
- SNOMED CT and LOINC give each data element an unambiguous meaning.
- EHDS makes structured Patient Summary exchange mandatory from March 2029.



Point-of-capture meaning is the enabling condition for display, sharing, reuse, and analysis.



coordinated delivery

Purpose: a uniform assessment framework

The document turns national policy and EHDS obligations into testable implementation expectations.

- Defines functional and technical requirements for SNOMED CT implementation.
- Aligns vendor solutions with BIHR, Belgian objectives, and EHDS readiness.
- Provides objective criteria for maturity assessment and registration.

**Requirements
specification**

**Objective
assessment**

**Snomedization
registration**

National scope is broader than cross-border exchange: SNOMED CT supports Belgian care, BIHR reuse, and secondary use.

**Belgian authorities
registration**

**EHDS March 2029
Patient Summary**

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SCT implementation - 4 Strategic Objectives

1. Terminology Provisioning
2. Data Entry
3. Storage
4. Interoperability

Implementation Categories

(cfr. SNOMED International)

- Terminology Services Provisioning
 - Versioning
 - Maintenance
- Data Entry
 - User Interface
 - Terminology Guidance
- Storage
 - Storage
 - Analytics & Reporting
- Interoperability
 - Interoperability

Objective 1

Terminology Services & Provisioning

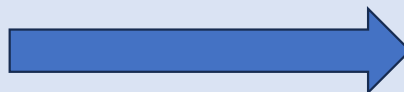
The eHR must include Terminology

→ Services



FHIR enabled services

→ Provisioning



Use latest Belgian Edition
Maintain subsets, maps and
translations

Objective 1: Terminology Services

N°	TSP1
Summary	FHIR enabled services
Description	The eHR shall provide terminology services conformant to the HL7 FHIR Terminology Service specification, implementing at minimum following operations against the SNOMED CT code system. ☐ \$lookup ☐ \$validate-code ☐ \$expand Applicable RefSets: ☐ Problems ☐ Observation ☐ Allergy & Intolerance ☐ Vaccination ☐ Procedures
Category	/
Deadline	Q4/26

Objective 1: Terminology Services

N°	TSP2
Summary	Semantic definition and evaluation of concept sets
Description	The FHIR terminology service shall support the dynamic definition and evaluation of SNOMED CT concept sets using the SNOMED CT hierarchy and relationships, including through ValueSet/\$expand and \$validate-code, without requiring manual enumeration or maintenance of all member concepts. The solution should support SNOMED CT ECL as the standard constraint mechanism; where ECL is not supported, the supplier shall demonstrate equivalent dynamic functionality for these operations.
Category	/
Deadline	Q4/26

Objective 1: Terminology Provisioning

N°	TSP3
Summary	Belgian Edition Release Currency
Description	Use a SNOMED CT release that is no more than one month behind the latest Belgian Edition.
Category	Versioning
Deadline	Q4/26

Objective 1: Terminology Provisioning

N°	TSP4
Summary	Controlled SNOMED CT Release Deployment
Description	Deploy new Belgian Edition releases without loss or corruption of concepts, descriptions, relationships, or reference sets, while minimizing service disruption.
Category	Maintenance
Deadline	Q4/26

Objective 1: Terminology Provisioning

N°	TSP5
Summary	Version consistency
Description	Use the same SNOMED CT version across all terminology-dependent components, or demonstrate that different versions can exchange data without changing clinical meaning.
Category	Versioning
Deadline	Q2/27

Objective 1: Terminology Provisioning

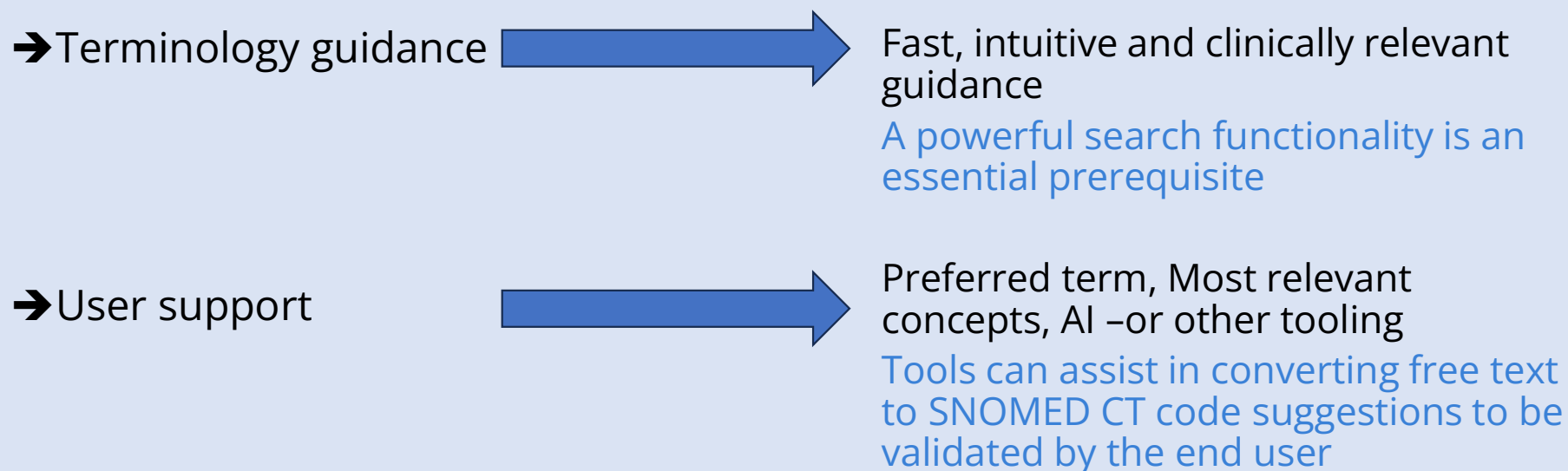
N°	TSP6
Summary	Multilanguage accommodation
Description	Display SNOMED CT content using the appropriate Belgian language reference sets (Dutch, French, German).
Category	Maintenance
Deadline	Q2/27

Objective 1: Terminology Provisioning

N°	TSP7
Summary	Terminology feedback mechanism
Description	The solution shall provide an accessible mechanism for users to report suspected missing or erroneous SNOMED CT content, including concepts, descriptions, translations, and mappings. Feedback shall ultimately be submitted through the official Belgian NRC feedback channel, either directly by the user or via the supplier as an intermediary.
Category	Versioning
Deadline	Q2/28

Objective 2 Data Entry

Data entry is made as easy as possible:



Objective 2: Terminology Guidance

N°	U01
Summary	Overview of set of basic search functionalities
Description	Provide efficient SNOMED CT search capabilities that help users quickly find and select the correct concept.
Category	Terminology Guidance
Deadline	Q2/27

SNOMED CT search and retrieval

Fast, tolerant and context-sensitive concept retrieval while the clinician types

≤ 1.0 s
initial results

$\geq 95\%$
of searches

Measure from the user interface action to the first displayed results. Include debounce and every other intentional delay under normal operating conditions.

01 Input behaviour

Progressive matching

as the user types characters into a SNOMED CT coded data element, suggest a list of relevant concepts but limit the selection of concepts

Implementation guidance

A supplier may use a minimum character threshold and debounce. Demonstration implementations use 3 characters and about 500 ms.

02 Matching logic

Description matching

A selectable concept must have a preferred or acceptable description that matches the entered text.

Tolerant search

Recognise plural and feminine forms, minor spelling errors, and compounds through word segmentation.

Multi-prefix, any order

Match each token as a whole word or word prefix, regardless of token order.

03 Scope and selection

Context first

Use the applicable terminology binding, language reference set or specialty subset to prioritise or restrict results.

Controlled fallback

If the context does not contain a suitable concept, allow a broader SNOMED CT search when the binding permits it.

Objective 2: Terminology Guidance

N°	U02
Summary	Search by Concept ID
Description	The solution shall support searching for SNOMED CT concepts by Concept ID and return the corresponding concept with an appropriate description based on the user's applicable language configuration. The solution should additionally support searching by Description ID and return the concept associated with that description.
Category	Terminology Guidance
Deadline	Q2/28

Objective 2: Terminology Guidance

N°	U03
Summary	Browsing through hierarchical relationships
Description	The solution shall enable users to explore the SNOMED CT relationships of a selected concept, including its parent and child concepts and other defined relationships. For each relationship, the solution shall identify the relationship type and related concept and allow navigation to that concept; the supplier may determine the appropriate user-interface mechanism for presenting these relationships.
Category	Terminology Guidance
Deadline	Q2/28

Objective 2: Terminology Guidance

N°	U04
Summary	Binding eHR with SNOMED CT
Description	The solution shall restrict SNOMED CT concepts available for search and selection to those permitted by the terminology binding associated with the relevant data element. The binding may be defined by a hierarchy, reference set, value set, ECL constraint, or other explicitly defined concept set; concepts outside that binding shall not be selectable.
Category	Terminology Guidance
Deadline	Q2/27

Objective 2: Terminology Guidance

N°	U05
Summary	Allow data capture via free text in case of missing concepts
Description	Where no suitable SNOMED CT concept can be identified for a SNOMED CT-coded data element, the solution shall allow the information to be recorded temporarily as free text. The free text shall be clearly distinguishable from SNOMED CT-coded data and shall not use any SNOMED CT Concept ID or placeholder code; the missing concept shall be reported through the feedback mechanism defined in TSP7.
Category	Terminology Guidance
Deadline	Q2/28

Objective 2: User Interface

N°	U06
Summary	Show user preferred display term for each concept.
Description	When a user selects a SNOMED CT concept, the solution shall display the value using the same term presented and selected during the search. The selected term shall not be automatically replaced by another description of the same concept during the current data-entry interaction.
Category	User Interface
Deadline	Q2/28

Objective 2: User Interface

N°	U07
Summary	Display of the most relevant concepts
Description	The solution shall provide at least one mechanism for rapidly accessing commonly used, recently used, personalized, or contextually relevant SNOMED CT concepts without performing a full terminology search. Concepts provided through such mechanisms shall meet the same validation requirements as standard search results, including active status and the terminology binding applicable to the data element.
Category	User Interface
Deadline	Q2/28

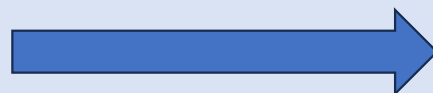
Objective 2: User Interface

N°	U08
Summary	Integration tools/solutions facilitating data search & capture
Description	The eHR shall support integration with tools or services that analyse clinical information and suggest candidate SNOMED CT concepts during data entry. Suggested concepts shall not be recorded without explicit user confirmation and shall meet the same validation requirements as standard selections, including active status and applicable terminology bindings; each suggestion shall provide the Concept ID and an appropriate description.
Category	User Interface
Deadline	Q2/28

Objective 3 Storage

Data shall be captured at the source using the semantic standard code within the eHR

→ Storage

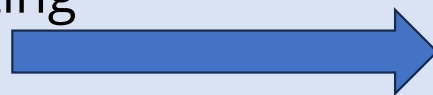


Store SNOMED CT Concept IDs

Preserve semantics and historical data

Status handling

→ Analytics & Reporting



Relations between concepts, concepts with “container” function, grouping of concepts, etc.

Objective 3: Storage

N°	S01
Summary	Store SNOMED CT concept identifiers in health records
Description	For every SNOMED CT-coded data element, the solution shall persist both the SNOMED CT Concept ID representing the recorded clinical meaning and the term displayed to or selected by the user at the time of recording. The Concept ID shall remain the authoritative coded representation; storing only a Description ID or free-text term instead of the Concept ID shall be considered a critical failure.
Category	Storage
Deadline	Q2/27

Objective 3: Storage

N°	S02
Summary	Clear representation of each concept ID of expression
Description	For each SNOMED CT-coded data element, the solution shall record a SNOMED CT concept representing the clinical meaning of the information being captured. Qualifiers, status indicators, workflow states, or other contextual attributes may supplement this concept but shall not replace it; the recorded SNOMED CT content together with the applicable data model shall make the intended clinical meaning unambiguous without reliance on free text.
Category	Storage
Deadline	Q2/27

Objective 3: Storage

N°	S03
Summary	Active SNOMED CT concepts for data entry
Description	The eHR shall allow only SNOMED CT concepts that are active in the applicable SNOMED CT version and permitted by the relevant terminology binding to be newly selected or recorded. Inactive concepts shall not be accepted as coded values for new clinical data.
Category	Storage
Deadline	Q2/27

Objective 3: Storage

N°	S04
Summary	Preservation of historically coded data
Description	If a recorded SNOMED CT concept is later inactivated, the eHR shall retain the originally recorded Concept ID unchanged as part of the historical clinical record. Any active replacement or successor concept shall be associated separately and shall not overwrite or modify the original Concept ID.
Category	Storage
Deadline	Q2/27

Objective 3: Analytics & Reporting

N°	S05
Summary	Use maps from SNOMED CT to national and international classifications
Description	The solution shall support officially available Belgian mappings between SNOMED CT and relevant classifications and terminologies, including ICD-10, ICPC-2, ICHI, NIHDI nomenclature, LOINC, and Orphanet, and shall derive target codes in accordance with the applicable mapping rules and metadata. Mapped results shall be available to relevant solution components, and the supplier shall document the supported mapping versions and how they are kept aligned with the corresponding Belgian SNOMED CT Edition or other official Belgian source.
Category	Analytics & Reporting
Deadline	Q2/28

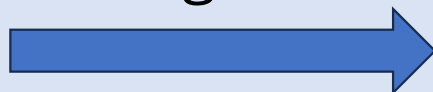
Objective 3: Analytics & Reporting

N°	S06
Summary	Ontology-Based querying
Description	For reporting, data extraction, and clinical decision support, the solution shall support evaluating SNOMED CT-coded data using the SNOMED CT ontology rather than requiring individual concept enumeration. Evaluation shall support both hierarchical is-a and relevant defining attribute relationships and shall reflect the SNOMED CT edition and version applicable to the processed data.
Category	Analytics & Reporting
Deadline	Q2/28

Objective 4 Interoperability

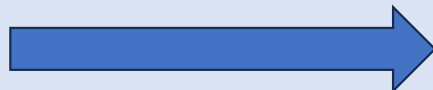
Data entered into the system shall be reusable for data exchange

→ Exchange



FHIR

→ Reusable data



One-time registration, multiple use

Objective 4: Interoperability

N°	I01
Summary	Use the national exchange standard FHIR to share SNOMED CT coded data in the eHR
Description	Use the national exchange standard FHIR to share data that are SNOMED CT coded in the eHR.
Category	Interoperability
Deadline	Q2/27

Objective 4: Interoperability

N°	I02
Summary	Reuse SNOMED CT coded data elements for all relevant message exchanges
Description	Reuse SNOMED CT coded data elements to (automatically) populate FHIR resources for all relevant message exchanges.
Category	Interoperability
Deadline	Q2/27

• Roadmap & Priorities

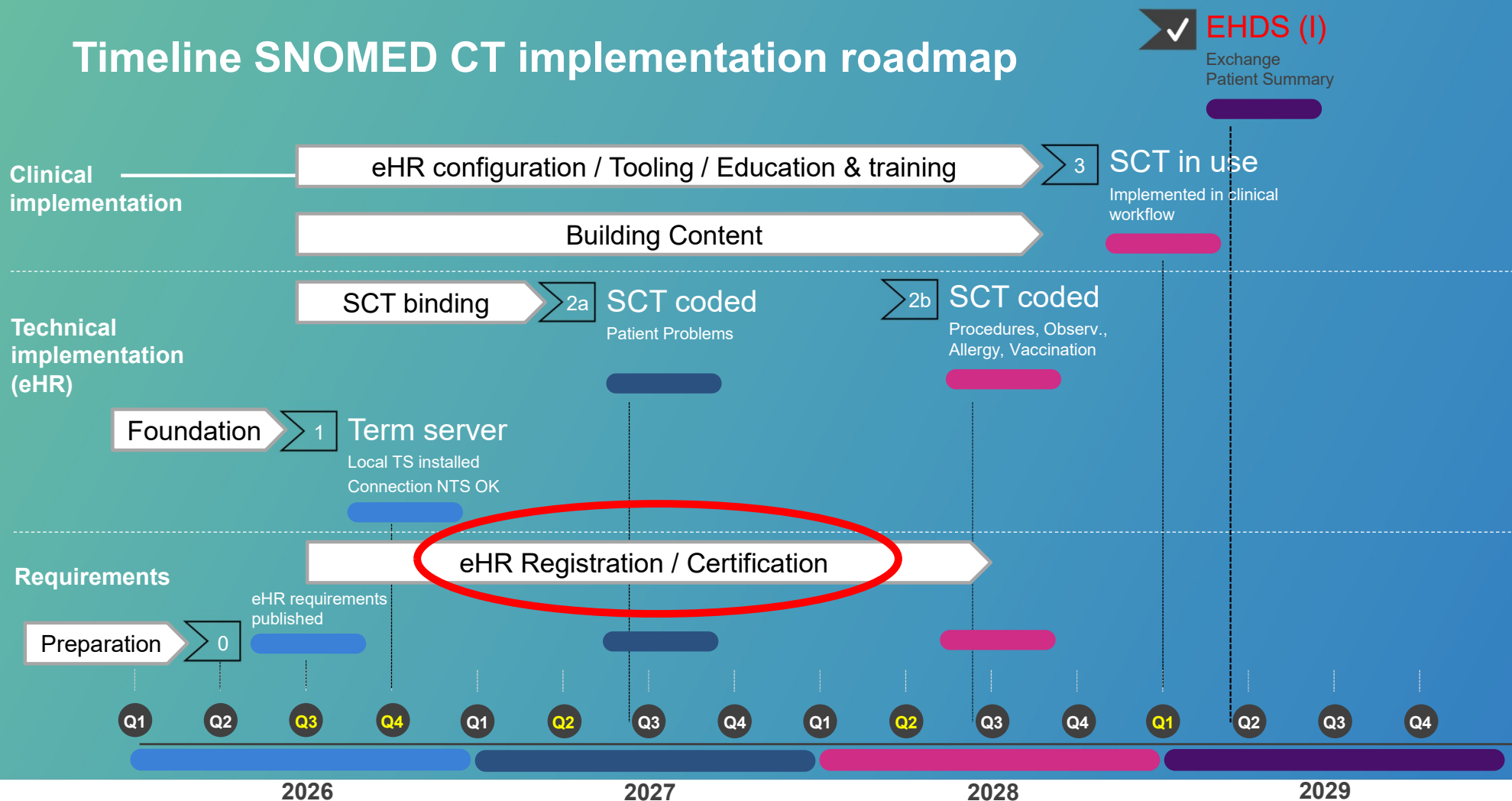
- Overview in excel
- # 23 requirements
- Phase implementation
- Date implementation & testing
- MoSCoW
 - Must
 - Should
- Github

Seq.	Req_ID	Requirement description	Objective	Category	MoSCoW	Phase impl.	Date impl.	Date testing
REQ01	TSP1	FHIR enabled services	Term. Services & Provisioning		Must	1. LTS	Q4/26	Q3/27
REQ02	TSP2	Semantic definition and evaluation of SNOMED CT concept sets	Term. Services & Provisioning		Must	1. LTS	Q4/26	Q3/27
REQ03	TSP3	Belgian Edition Release Currency	Term. Services & Provisioning	Versioning	Must	1. LTS	Q4/26	Q3/27
REQ04	TSP4	Controlled SNOMED CT Release Deployment	Term. Services & Provisioning	Versioning	Must	1. LTS	Q4/26	Q3/27
REQ05	TSP5	Version Consistency	Term. Services & Provisioning	Versioning	Must	2a - Problem	Q2/27	Q3/27
REQ06	TSP6	Multilanguage accommodation	Term. Services & Provisioning	Maintenance	Must	2a - Problem	Q2/27	Q3/27
REQ07	TSP7	Provide a reporting/feedback mechanism	Term. Services & Provisioning	Versioning	Should	2b. Patient Summary	Q2/28	Q3/28
REQ08	U01	Overview of set of basic search functionalities	Data Entry	Term. Guidance	Must	2a - Problem	Q2/27	Q3/27
REQ09	U02	The system will enable searches using Concept ID	Data Entry	Term. Guidance	Should	2b. Patient Summary	Q2/28	Q3/28
REQ10	U03	Following the return of search results, the system will permit users to browse through nearby hierarchy	Data Entry	Term. Guidance	Should	2b. Patient Summary	Q2/28	Q3/28
REQ11	U04	Limit searching and selection to only those concepts from SNOMED CT that have been specifically bound to that data element	Data Entry	Term. Guidance	Must	2a - Problem	Q2/27	Q3/27
REQ12	U05	Allow data capture via free text when concepts cannot be found	Data Entry	Term. Guidance	Should	2b. Patient Summary	Q2/28	Q3/28
REQ13	U06	Show user preferred display term for each SNOMED CT concept	Data Entry	User Interface	Should	2b. Patient Summary	Q2/28	Q3/28
REQ14	U07	The eHR system shall display the most relevant concepts	Data Entry	User Interface	Should	2b. Patient Summary	Q2/28	Q3/28
REQ15	U08	The eHR is capable of integrating tools/solutions facilitating the search and capture of clinical data	Data Entry	User Interface	Should	2b. Patient Summary	Q2/28	Q3/28
REQ16	S01	The eHR must store SNOMED CT concept identifiers in the health records	Storage	Storage	Must	2a - Problem	Q2/27	Q3/27
REQ17	S02	Ensure that the context of each SNOMED CT concept identifier or expression is clearly represented	Storage	Storage	Must	2a - Problem	Q2/27	Q3/27
REQ18	S03	Active SNOMED CT concepts for data entry	Storage	Storage	Must	2a - Problem	Q2/27	Q3/27
REQ19	S04	Preservation of historically coded data	Storage	Storage	Must	2a - Problem	Q2/27	Q3/27
REQ20	S05	Use maps from SNOMED CT to national and international classifications including legacy and current data	Storage	Analytics and Reporting	Should	2b. Patient Summary	Q2/28	Q3/28
REQ21	S06	Use SNOMED CT's defining relationships in the determination of the correct results or outcomes	Storage	Analytics and Reporting	Should	2b. Patient Summary	Q2/28	Q3/28
REQ22	I01	Use the national exchange standard FHIR to share data that are SNOMED CT coded in the eHR	Interoperability	Interoperability	Must	2a - Problem	Q2/27	Q3/27
REQ23	I02	Reuse SNOMED CT coded data elements for all relevant message exchanges	Interoperability	Interoperability	Must	2a - Problem	Q2/27	Q3/27

AGENDA

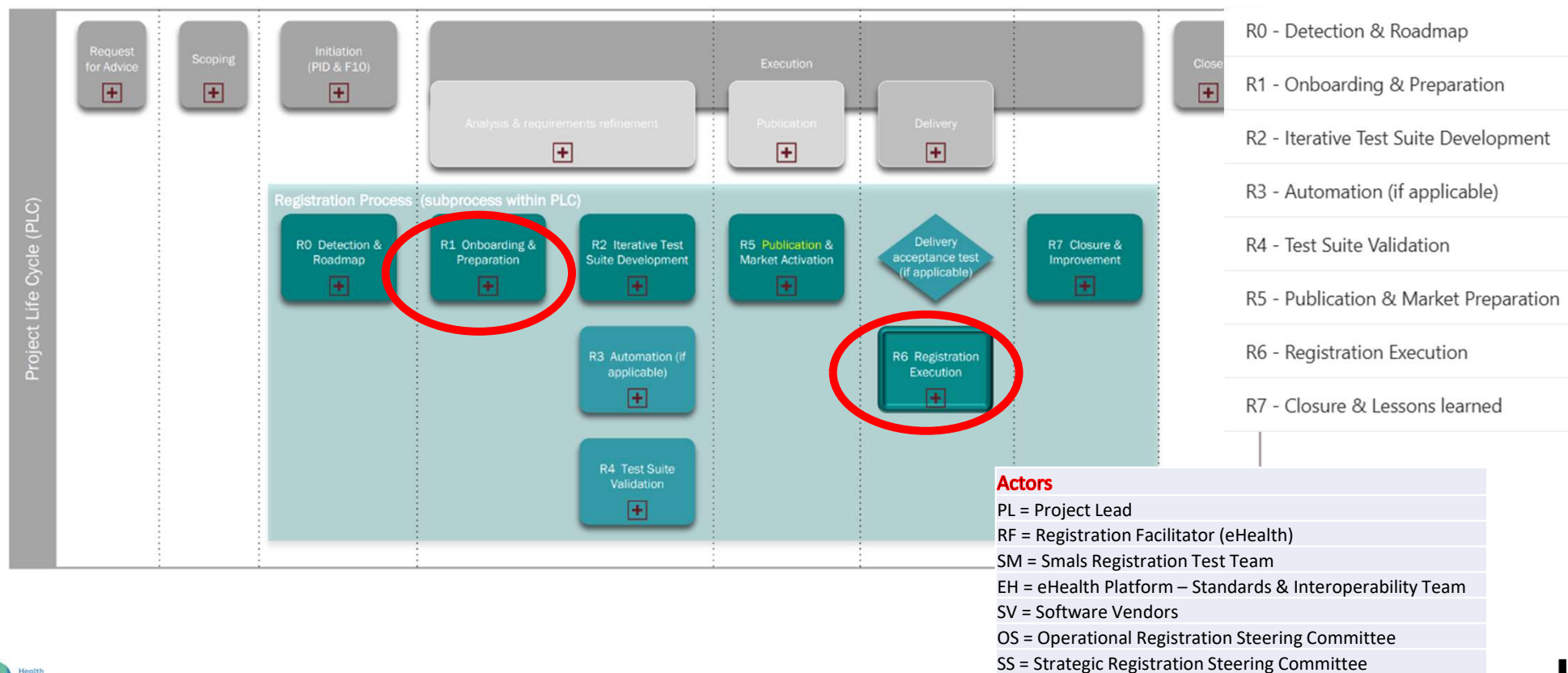
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Timeline SNOMED CT implementation roadmap



Registration - Certification

Project Life Cycle & Registration Process – Reference Overview



Registration process

Responsable: **Software vendors**

- › [Wat is registratie?](#)
- › [Wie is verantwoordelijk voor de registratie van software?](#)
- › [Waarom uw software registreren?](#)
- › [Welke criteria zijn van toepassing?](#)
- › [Registratie voor een specifieke digitale toepassing](#)
- › [Hoe schrijft u zich in voor een registratie?](#)
- › [Vragen?](#)
- › [Contacten](#)
- › [Qu'est-ce que l'enregistrement ?](#)
- › [Qui est responsable de l'enregistrement d'un logiciel ?](#)
- › [Pourquoi enregistrer votre logiciel ?](#)
- › [Quels critères s'appliquent ?](#)
- › [Enregistrement pour une application numérique spécifique](#)
- › [Comment s'inscrire à un test d'enregistrement?](#)
- › [Des questions?](#)
- › [Contacts](#)

<https://www.inami.fgov.be/fr/themes/esante/informations-techniques-et-enregistrement-pour-les-fournisseurs-de-logiciels-destines-aux-dispensateurs-de-soins>

<https://www.riziv.fgov.be/nl/thema-s/egezondheid/technische-informatie-en-registratie-voor-softwareleveranciers-van-softwarepakketten-voor-zorgverleners>

Registration – levels

1. Registration for a Specific Digital Application

- Examples: SAM v2, Digital Referral Prescription, Clinical Decision Support Platform
- Focused on a single integration
- > **SNOMED CT implementation**

2. Full Certification Path

- Designed for a specific healthcare provider group
- Covers both functional and modular criteria

Which criteria?

- **Modular Criteria**

- Requirements specific to a digital application
- Vary according to the integrated services and projects
- > **SNOMED CT implementation – Requirements**

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Github – sct_implementation

- Goal

Provide a central location for collecting and sharing all relevant documentation, issues, examples, and related resources, ensuring that all information is accessible and maintained in one place.

- Who ?

- all stakeholders

Software vendors – healthcare professionals – terminology experts

Analysts – Projectmanagers – Testers - ...

- Why?

- Feedback & communication is valuable
 - Overall quality and consistency of the SNOMED CT implementation
 - Identify & solve issues

Structure

00_TIMELINE
01_Requirements Specification Document
02_Requirements Roadmap & Milestones
03_Clinical Requirements
03_Technical Requirements
04_Registration
README.md

https://github.com/ehealthplatformstandards/sct_implementation

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- **Varia**

LTS – Audit

- First session - > Wednesday 30/09 & Thursday 01/10
- Second session -> end November / begin December
- Online – 1 Hour
- After reservation -> detailed testing instruction
 - [Testplan](#)
- **Timeslots available !**

Questions?

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Thanks!